

Charger being an important part of the mobile computing world, what should the designers keep in mind whilst designing a charger?

People have many devices with a big or small screen, lot of processing power and lot of memory. Some devices also have big batteries but people do not want to wait long to get them charged. They need to be charged quickly. So we need a very high-current charger but it cannot be very big because we need to carry it in our pocket or in our bag. It has to be smaller than a laptop charger but bigger than a small phone charger. So designing such a charger is not an easy task, it has to be custom designed. A custom-designed charger is a charger that can withstand electrical transients. The fuse inside the charger should open in a safe way and should not have any loud popping sound, any smoke or black soot coming up. At the same time, the fuse needs to open up very fast and also very safely so that it does not cause any issue. It should meet the requirements of the electrical transient tests and short-circuit tests or interrupt current test, and these tests should counterbalance each other. The charger needs to be fine tuned with the product it is made for, and also requires very close customer engagement and several iterations of testings in the lab to get it right.

What options are available to an engineer towards protecting the batteries inside the mobile phones, laptops, tablets, etc?

The mobile phones, laptops, tablets, etc have Li-ion batteries, which have a very dangerous chemistry yet are lighter rechargeable batteries. The Li-ion battery pack design has 2-3 main levels of protection. The first one is the main semiconductor-based IC called battery management unit (BMU). The BMU is responsible for managing the charging and discharging of the battery and it also provides the primary protection. So if the battery draws too high a current, it would turn the pack off.

The secondary protection is either a fuse or a resettable fuse (PTC). The PTC works in a very interesting way, very close to the battery pack. When the battery pack starts to heat up, it opens the PTC, which is a temperature-sensitive device with positive temperature coefficient. As temperature increases, PTC's resistance increases and turns the pack off. After the fault is removed and the battery pack cools down, then PTC resets itself and makes the battery pack operational again. Over the years, the use of the PTC and the Li-ion battery pack has historically become very popular because of its protection method.

What is SIDACTor?

It came from the thyristor family of devices—SCRs, quadracs, triacs and SIDACs. It is also a lightening protector used especially for telecommunication light applications like DSL, broadband and copper. It is a silicon device based on thyristor technology and, depending upon the voltage that is supplied to the device, it allows that device to turn on

and off. When SIDACTor turns on, it allows the lightening waveform to pass through it to the ground and then it turns off when the lightening subsides. It is optimised for very sensitive lightening.

How can the protection level of the devices be decided by the designers?

The devices are more related to the safety and the standards. The primary AC line is considered to be very dangerous, and it takes a very long time to get activated and deactivated when the consumer is in its direct vicinity. There is no particular extent or level of protection for such devices; they need to be permanently disconnected. These need to pass the certification by a compliance officer, to show the product number and the certification. On the secondary side of the circuit we have where a PTC can be used.

Are there any kind of circuit protections available for the high-fidelity audio systems?

The high-fidelity or high-end needs of the audio systems require electrostatic discharge (ESD) protection. Speakers can get overdriven. If the amplifier puts in a lot of power into the speakers, the speakers would be destroyed. So a PTC is placed in the power section of the speakers. As the current going into the speaker builds up, the resistance in the PTC increases. And once it goes beyond a certain point, the PTC introduces high-end resistance and thus draws the current down. When the volume is turned down, the PTC turns the resistance down and the stereo comes back to normal operation. This is a very old application.

What are GFCI and AFCI?

GFCI is the ground fault current interrupt and AFCI is the arc fault current interrupt. Lot of times GFCI has been turned to AFCI, but our applications are the same in both the cases. They use varistors, metal oxide varistors and over-surge products. Since these are safety devices, they have to be reliable, because when they are needed, they cannot fail, and they have to work for many many years. Now during this long duration, lot of electrical transients or surges would come into the GFCI or AFCI, so they put a varistor inside them, that clamps in anything that tries to get in there and attempts to damage the product. It is a very focussed segment and we sell lots of varistors in that segment.

We have started a lot of research in the AFCI to find out if there are any specific standards that would give us an advantage. The protection in these areas is usually single-stage as it is pretty low-cost and small-scale.

What are the variations that the designers would face whilst they are reducing the size of the product?

Yes, it is a compromise. Whenever you shrink the product, you decrease its capability to handle the amount of surge or amount of current. So if we take the example of a fuse, the smaller it is, the less capable it would be for large faults. ●